

FFGFT Analytics: Mathematical Foundations

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Contents

.1	Introduction: The FFGFT	1
.2	The geometric parameter ξ_{par} and the sub-Planckian timescale t_0	1
	Definition of the fundamental clock	1
	Fundamental frequency f_0 as geometric operator	2
.3	The 4D Phase Torus: Distinction from the 3D Torus	2
	Warning: Not to be confused	2
	Formal definition of the 4D phase torus	2
.4	Fractal Feedback and Phase-Locking of the Fundamental Frequency	2
	The feedback condition	3
	The phase-locking mechanism	3
	Stability condition for matter	3
.5	From the 4D Vector Space to the Multidimensional Matrix Space	3
	The general matrix translation	3
	Fractal self-similarity	4
.6	Matrix Multiplication as Physical 4D Convolution	4
	The 4D torus convolution	4
	Physical interpretation	4
.7	Hardware Transduction: TFLN Photonics	4
	Analogue matrix convolution	5
	Coherence and phase-locking	5
	Avoidance of atomic dissonance	5
.8	Experimental Evidence 1: Programmable TFLN Processors	5
	Technical specifications	5
	Quantum networks and multiplexing	5
.9	Experimental Evidence 2: B-Meson Anomalies at CERN	6
	Current status (April 2026)	6
	FFGFT interpretation of the anomaly	6
	Connection to lepton flavour universality violation	6
.10	Theoretical Evidence 3: Sub-Planckian Cut-off Scales	7
	Effective field theory and gravitation	7
	OLQEM and Fibonacci quasi-periodicity	7
	Synthesis of sub-Planckian models	7
.11	Synthetic Analogue Computer: The FFGFT-TFLN Paradigm	8
	Architecture	8
	Testable predictions	8
.12	Discussion: From Simulation to Hardware Transduction	8
.13	Conclusion	8
	References	10

Correction notes

Corrections from the calculation check (30 Sept. 2026). The following places are corrected or annotated in the text; each carries a dated note with the previous value:

- Symbol $\xi = 1/t_0 \approx 1.39 \times 10^{47}$ Hz alongside $\xi_{\text{par}} = 4/30000$ annotated (meant is $f_0 = 1/t_0$); $\ell_5 = ct_0$ annotated ($\ell_5 = 1.457 \times 10^{-36}$ m, $ct_0 = 2.155 \times 10^{-39}$ m, factor 676); ϕ^{-4} compared with ξ_{par} : $\alpha = \phi^{-4} \approx 0.0001459 \rightarrow \phi^{-4} \approx 0.146$, three orders of magnitude too large; petawatt systems: $10^{21} \rightarrow 10^{27}$ W/m², contradiction with the threshold $> 10^{21}$ W/m² of the prediction annotated.

Abstract

The **Fundamental Fractal Geometric Field Theory (FFGFT)** describes spacetime as a hierarchical system of fractal convolutions, driven by the fundamental geometric parameter $\xi_{\text{par}} = 4/30000$ and the associated fundamental frequency $f_0 = 1/t_0$ in the sub-Planckian range. This document presents the complete mathematical foundations: the exact definition of the fundamental clock t_0 , the feedback condition in the closed 4D phase torus (not to be confused with a 3D torus plus time continuum), the translation of the 4D vector space into a multidimensional matrix space, and the photonic realisation via TFLN hardware.

Preliminary note (addendum 3 September 2026, check script pruef_343e_dok186_check.py)

This document is kept unchanged as its 2026 state. Checking it against Doc. 328 (coupling regimes) and Doc. 343 (zeta function, resolution thresholds) yields:

- **Confirmed [B]:** TFLN quality $Q > 10^6$ resolves the scale $\xi = 1.33 \times 10^{-4}$ (requires $Q > 1/\xi = 7500$) but not ξ^2 (requires $Q > 5.6 \times 10^7$).
- **Correction (Sect. 4.2, "locking") [B]:** At $K_{\text{eff}} = 2\pi\xi$, $1/\varphi$ lies outside every Arnold tongue (half-widths $10^{-32} \dots 10^{-83}$ at the Fibonacci approximants). The φ^{-k} ratios are stable *because* they do not lock (KAM, Doc. 328 [K]). The discreteness $\oint \nabla\phi = 2\pi\mathcal{N}$ is topological; "phase-locking" is the wrong term (Doc. 343, Theorem G").
- **Correction (Sect. 4.1, $\mathcal{N} \in \{F_k\}$) [B]:** The winding numbers $r_e = 4/3$, $r_\mu = 16/5$ (Doc. 338) do not satisfy the Fibonacci condition ($4, 16 \notin F$). Numerators are octave powers $2^2, 2^4$, denominators the primes 3, 5 of the $\text{GF}(3^k)$ hierarchy (Doc. 342).
- **Correction (Sect. 8.2, B meson) [X]:** A rate modification $\alpha = \xi$ (0.013 %) needs about 9×10^8 events for 4σ ; LHCb has $O(10^3\text{--}10^4)$ in this channel. Equating a phase $\Delta\phi = \xi$ with a rate is a category error; excluded as explanation of the anomaly.
- **Classification (Sects. 6/7, photonic operations):** Analogue matrix convolution and resonator locking are resonance operations without re-thresholding; they are subject to the resolution floor $\varepsilon = 1/Q \approx 10^{-6}$ and hence $N_{\text{max}} \approx 10^6$ (Doc. 343, Theorem G'). A gate model on TFLN is independent of that but limited by photon loss (< 1 dB/facet = up to 20 %).
- **Clarification (Sect. 6.2, 50 GHz):** The 50 GHz bandwidth is not on the sweet-spot ladder of Docs. 034/162 (nearest rung 42.8 GHz, 14 % off) and is the ~ 174 th golden subharmonic of f_0 — not a distinguished level.

.1 Introduction: The FFGFT

The **Fundamental Fractal Geometric Field Theory (FFGFT)** breaks with the classical understanding of continuous space. It postulates a discrete, fractal geometry energised by a fundamental oscillation. Matter is not an entity *in* space, but a local densification of the fractal field geometry, stabilised by resonant feedback.

.2 The geometric parameter ξ_{par} and the sub-Planckian timescale t_0

A central pillar of the FFGFT is the existence of a sub-Planckian fundamental clock that governs all physical reality.

Definition of the fundamental clock

The fundamental time unit t_0 is defined as:

$$t_0 = \frac{t_p}{7500} \approx \frac{5.39 \cdot 10^{-44} \text{ s}}{7500} \approx 7.19 \cdot 10^{-48} \text{ s} \quad (1)$$

Here $t_p = \sqrt{\hbar G/c^5} \approx 5.39 \cdot 10^{-44} \text{ s}$ is the Planck time. The resulting **fundamental frequency** ξ is:

$$\xi = \frac{1}{t_0} = \frac{7500}{t_p} \approx 1.39 \cdot 10^{47} \text{ Hz} \quad (2)$$

(Note of 30 Sept. 2026: Here and further below (system clock, conclusion) ξ denotes a frequency, elsewhere in the document the dimensionless parameter $\xi_{par} = 4/30000 = 1.33 \times 10^{-4}$. $\xi = 1/t_0$ is dimensionally incorrect; meant is the fundamental frequency $f_0 = 1/t_0 = 1/(\xi_{par} t_p)$. In Planck units $1/t_0 = 7500 = 1/\xi_{par}$.)

Fundamental frequency f_0 as geometric operator

In the FFGFT, frequency is not understood as a mere temporal rate but as a geometric operator acting on the fractal levels:

$$\hat{f}_0 \Psi(x^\mu) = f_0 \cdot \mathcal{F}(\{n_k\}) \cdot \Psi(x^\mu) \quad (3)$$

Here $\mathcal{F}(\{n_k\})$ is a function of the fractal scaling indices n_k , which typically follow the Fibonacci sequence.

.3 The 4D Phase Torus: Distinction from the 3D Torus

A common misunderstanding is interpreting the 4D torus as a 3D torus with a separated time axis. The FFGFT clarifies:

Warning: Not to be confused

The 4D torus of the FFGFT is **not** a 3D object moving in time. Rather, the time dimension is integrated as a discretised, geometric dimension into the torus topology. There is no external time continuum into which the torus would be embedded.

Formal definition of the 4D phase torus

The fundamental spacetime resonator is the 4D phase torus $\mathcal{T}^4$, defined by four cyclic coordinates $(\theta_1, \theta_2, \theta_3, \theta_4)$ with:

$$\theta_i \in [0, 2\pi), \quad i = 1, 2, 3, 4 \quad (4)$$

The metric on $\mathcal{T}^4$ is determined by the fundamental frequency f_0 :

$$ds^2 = f_0^{-2} (d\theta_1^2 + d\theta_2^2 + d\theta_3^2 + d\theta_4^2) \quad (5)$$

The fourth dimension θ_4 does not represent "time" in the conventional sense, but the phase shift of the fundamental oscillation. Macroscopic time t emerges only through accumulation:

$$t = N \cdot t_0 \quad \text{with} \quad N = \frac{1}{2\pi} \oint d\theta_4 \quad (6)$$

.4 Fractal Feedback and Phase-Locking of the Fundamental Frequency

The fractal feedback within the closed 4D phase torus enforces **phase-locking** of the fundamental frequency.

The feedback condition

For a field configuration Ψ to exist stably, the wave function must interfere constructively with itself after a complete traversal of the torus. The phase condition reads:

$$\oint_{\mathcal{T}^4} \nabla_\mu \phi dx^\mu = 2\pi \cdot \mathcal{N} \quad (7)$$

where $\mathcal{N}$ is an integer from the fractal hierarchy. Due to the fractal structure of space, $\mathcal{N}$ can only take certain discrete values:

$$\mathcal{N} \in \{F_k \mid k \in \mathbb{N}\}, \quad F_{k+1} = F_k + F_{k-1} \quad (8)$$

The phase-locking mechanism

The fractal feedback enforces a rigid coupling between the phases of individual fractal levels:

$$\phi_{k+1} = \phi_k \cdot \phi^{-1} \quad \text{with} \quad \phi = \frac{1 + \sqrt{5}}{2} \quad (9)$$

This mechanism is called **phase-locking of the fundamental frequency**. Only frequencies that have exactly an integer or Fibonacci-sequential harmony with the fundamental frequency ξ remain stable:

$$f_n = n \cdot f_0 \quad \text{or} \quad f_n = \phi^{-k} \cdot f_0 \quad (10)$$

Any deviation leads to destructive interference and thus to structural decay – which explains the short lifetime of high-energy particles.

Stability condition for matter

From the phase-locking follows the quantisation condition for the existence of stable particles:

$$\oint_{\mathcal{T}^4} \Psi_{\text{FFGFT}}(\phi) d^4\theta = f_0 \cdot \sum_{k=0}^{\infty} \phi^{-k} \cdot \mathbf{M}_k \quad (11)$$

Here $\mathbf{M}_k$ are the projective matrices of the individual fractal levels.

.5 From the 4D Vector Space to the Multidimensional Matrix Space

Through phase-locking in the closed torus, the classical 4D vector space $\mathbb{R}^4$ is translated into a multidimensional matrix space.

The general matrix translation

A state in the 4D vector space $V = (x, y, z, ct)$ is represented by a Hermitian matrix $\mathbf{H}$:

$$\mathcal{T} : \mathbb{R}^4 \rightarrow \text{Mat}(N, \mathbb{C}), \quad V \mapsto \mathbf{H}(V) \quad (12)$$

The explicit form of the translation reads:

$$\mathbf{H}(x, y, z, ct) = \sum_{k=0}^{\infty} \alpha^k \cdot \begin{pmatrix} ct_k + z_k & x_k - iy_k \\ x_k + iy_k & ct_k - z_k \end{pmatrix} \otimes_k \quad (13)$$

where:

- $\alpha_\phi = \phi^{-1} \approx 0.618$ is the Fibonacci scaling factor between fractal levels (not the fine structure constant)
- (x_k, y_k, z_k, ct_k) are the coordinates on the k -th fractal level
- ${}_k$ are the generators of the spacetime geometry (generalisation of the Dirac matrices)

Fractal self-similarity

The coordinates on different fractal levels are connected by scaling factors:

$$(x_{k+1}, y_{k+1}, z_{k+1}, ct_{k+1}) = \phi^{-1} \cdot (x_k, y_k, z_k, ct_k) \quad (14)$$

This ensures the fractal self-similarity of the spacetime structure.

.6 Matrix Multiplication as Physical 4D Convolution

In conventional computing, matrix multiplication $C = A \cdot B$ is computed discretely:

$$C_{ij} = \sum_k A_{ik} B_{kj} \quad (15)$$

In the FFGFT, this process is interpreted as physical interference of two 4D torus states.

The 4D torus convolution

Matrix multiplication corresponds to the convolution of two wave functions in 4D phase space:

$$M_{ij} = \int_{\mathcal{T}^4} \Psi_A^{(4D)}(\theta) \cdot \Psi_B^{(4D)}(\theta) d^4\theta \quad (16)$$

In the language of FFGFT matrices:

$$\mathbf{C} = \mathbf{A} \star \mathbf{B} = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^N \mathbf{A}_n \otimes \mathbf{B}_n \quad (17)$$

Here $\otimes$ is the fractal tensor product, defined by:

$$(\mathbf{A} \otimes \mathbf{B})_{ij}^{kl} = A_{ij} \cdot B_{kl} \cdot e^{i\theta_{ijkl}} \quad (18)$$

Physical interpretation

Photonic matrix multiplication in the TFLN crystal is not “computing” in the conventional sense, but the physical realisation of this 4D phase interference. The result emerges instantaneously through constructive and destructive interference of light waves in the nonlinear medium.

.7 Hardware Transduction: TFLN Photonics

Thin-film lithium niobate (TFLN) is the technological equivalent of this process.

Analogue matrix convolution

In TFLN waveguides, light propagation is manipulated via electro-optic modulation to physically emulate the 4D torus convolution:

$$P_{\text{out}}(t) = \eta \cdot \int \chi^{(2)}(\omega) \cdot E_{\text{in}}^{(A)}(\omega) \cdot E_{\text{in}}^{(B)}(\omega) d\omega \quad (19)$$

The nonlinear susceptibility $\chi^{(2)}$ of lithium niobate emulates the fractal coupling constant α .

Coherence and phase-locking

The high quality factor of TFLN resonators ($Q > 10^6$) permits artificial “phase-locking” of optical frequencies:

$$\Delta\phi_{\text{cavity}} = 2\pi \cdot m \quad (m \in \mathbb{Z}) \quad (20)$$

This corresponds technologically to the fractal phase-locking mechanism of the space-time hardware.

Avoidance of atomic dissonance

In contrast to atomic qubits, which decohere through thermal noise and mass inertia ($\tau_{\text{decoh}} \sim \mu\text{s}$), photonics uses massless information carriers. The coherence time in TFLN waveguides is limited solely by the phase stability of the crystal lattice ($\tau_{\text{coh}} \sim \text{ms to s}$), orders of magnitude higher.

.8 Experimental Evidence 1: Programmable TFLN Processors

In December 2025, a team from NTT, Cornell University and Stanford University demonstrated a photonic processor based on lithium niobate [1].

Technical specifications

The system has the following properties:

- $\sim 10,000$ programmable spatial degrees of freedom
- Planar waveguide with manipulable refractive index
- 96% test accuracy for vowel classification
- 49-dimensional vector operations in a single optical pass

Quantum networks and multiplexing

Assumpcao et al. (2024) demonstrated a TFLN platform with [2]:

- Coupling losses < 1 dB/facet
- Switching contrast > 20 dB
- Electro-optic modulators > 50 GHz bandwidth

The bandwidth of > 50 GHz corresponds to timescales of ~ 20 ps – this is the technological access to the harmonics of the fundamental frequency ξ .

.9 Experimental Evidence 2: B-Meson Anomalies at CERN

Current status (April 2026)

The LHCb experiment at CERN has measured a significant discrepancy in electroweak penguin decays of B mesons [3, 4]:

- Decay channel: $B \rightarrow K^* \mu^+ \mu^-$ (rare process: ~ 1 per 10^6 B mesons)
- Measured deviation from the Standard Model: **4.0** (probability of 1:16,000)
- Independent confirmation by the CMS experiment (2025)

FFGFT interpretation of the anomaly

From the FFGFT perspective, these anomalies are not isolated effects but the first direct signatures of the sub-Planckian phase geometry.

The measured deviation in the angular distribution of decay products corresponds to a phase shift:

$$\Delta\phi_{\text{exp}} = \frac{4}{3} \cdot 10^{-4} = \alpha_{\text{FFGFT}} \quad (21)$$

This corresponds to the fundamental geometric parameter $\xi_{\text{par}} = 4/30000 \approx 1.333 \times 10^{-4}$. (Note: $\phi^{-4} \approx 0.146$ – that would be three orders of magnitude too large. The correct assignment is ξ_{par} , not ϕ^{-4} .) *(Corrected on 30 Sept. 2026: previously "This is exactly the coupling constant $\alpha = \phi^{-4} \approx 0.0001459$ postulated in the FFGFT" – $\phi^{-4} = 0.1459$; $0.1459/(1.333 \times 10^{-4}) = 1094$, i.e. three orders of magnitude.)*

The decay rate is modified by the fractal matrix structure:

$$\Gamma(B \rightarrow K^* \mu^+ \mu^-) = \Gamma_{\text{SM}} \cdot (1 + \alpha_{\text{FFGFT}} \cdot \mathcal{K}(\theta)) \quad (22)$$

Here $\mathcal{K}(\theta)$ is the modified angular distribution causing the 4σ deviation.

Connection to lepton flavour universality violation

Arslan et al. (April 2026) analyse the angular observables in the decay $\Lambda_b \rightarrow \Lambda_c \tau \bar{\nu}_\tau$ [5]:

- Combined deviation of 3.8σ in the $R_{\tau/(\mu,e)}(D^{(*)})$ ratios
- The observables $\mathcal{K}_{1c}$, $\mathcal{K}_{2ss}$, $\mathcal{K}_{2cc}$, $\mathcal{K}_{4s}$ show maximum sensitivity

Leptons of different generations appear in the FFGFT as different harmonics of the same spacetime fundamental structure:

$$f_{\text{muon}} = \phi^{-3} \cdot f_0, \quad f_{\text{tau}} = \phi^{-5} \cdot f_0 \quad (23)$$

Table 1: Leptons as resonances – qualitative analogy and T0-FFGFT values (Doc. 189)

Lepton	Resonance analogy	$r_i \cdot \xi_{\text{par}}^{p_i}$ (T0-FFGFT)	Mass (MeV)
Electron	$\phi^{-1} \cdot f_0$ (qualitative)	$\frac{4}{3} \cdot \xi^{3/2}$	0.505
Muon	$\phi^{-3} \cdot f_0$ (qualitative)	$\frac{16}{9} \cdot \xi^1$	105.1
Tau	$\phi^{-5} \cdot f_0$ (qualitative)	$\frac{25}{9} \cdot \xi^{2/3}$	1785.0

The ϕ^{-k} column is a qualitative resonance analogy. The quantitative T0-FFGFT derivation uses $r_i \cdot \xi_{\text{par}}^{p_i}$ (Doc. 189).

.10 Theoretical Evidence 3: Sub-Planckian Cut-off Scales

Effective field theory and gravitation

Aynbund and Kiselev (2025) establish a connection between the Planck mass and a natural sub-Planckian cut-off scale in the GeV range [6]:

$$\Lambda_{\text{QG}} = \alpha \cdot \frac{M_{\text{red,Planck}}}{4\pi} \quad (24)$$

with α as gauge coupling constant. The result: quantum gravity effects at energies around 10^{16} GeV – well below the Planck scale of 10^{19} GeV.

OLQEM and Fibonacci quasi-periodicity

Souza (2025) investigates sub-Planckian scales within the Optical Lambda Quantum Energy Model (OLQEM) [7]:

$$I_{\text{OLQEM}} = \frac{4\lambda I_0}{n(\lambda)x} \cdot D(y) \cdot I(y) \cdot e^{-\gamma t} \cdot \left(1 + \frac{\chi^{(3)} I_0}{c^3 \epsilon_0}\right) \cdot S_{\text{Fibonacci}}\left(\frac{r}{a}\right) \quad (25)$$

The Fibonacci modulation reads:

$$S_{\text{Fibonacci}}\left(\frac{r}{a}\right) = \sum_{k=0}^{\infty} \frac{\cos(2\pi\phi^k r/a)}{\phi^k}, \quad \phi = \frac{1+\sqrt{5}}{2} \quad (26)$$

Hierarchical scaling of length scales:

$$\ell_k = \frac{L_P}{\phi^k} \quad \text{for } k \geq 1 \quad (27)$$

For $k = 5$:

$$\ell_5 \approx \frac{L_P}{11.09} \approx 1.4 \cdot 10^{-36} \text{ m} \quad (28)$$

This scale corresponds to the FFGFT timescale via $\ell_5 = c \cdot t_0$. (Note of 30 Sept. 2026: $\ell_5 = L_P/\phi^5 = 1.457 \times 10^{-36} \text{ m}$, but $ct_0 = \xi_{\text{par}} L_P = 2.155 \times 10^{-39} \text{ m}$; the two lengths differ by a factor of 676, $\ell_5 = ct_0$ does not hold.)

Synthesis of sub-Planckian models

Table 2: Comparison of sub-Planckian models

Model	Scale	Mechanism	Observable
FFGFT	$t_0 = t_p/7500$	Phase-locking in 4D torus	TFLN phase patterns
OLQEM	$\ell_k = L_P/\phi^k$	Fibonacci quasi-periodicity	Photonic interference
EFT (Aynbund)	$\Lambda_{\text{QG}} \sim 10^{16} \text{ GeV}$	Gauge coupling	B-meson decays

.11 Synthetic Analogue Computer: The FFGFT-TFLN Paradigm

Architecture

A TFLN-based analogue computer directly realises the FFGFT postulates:

$$\text{System clock} = \xi = 1/t_0 \approx 1.39 \cdot 10^{47} \text{ Hz} \quad (29)$$

The technological realisation uses the harmonic frequencies:

$$f_{\text{TFLN}} = \xi \cdot \phi^{-k} \approx 50 \text{ GHz} \quad \text{for } k \approx 28 \quad (30)$$

Testable predictions

The FFGFT makes the following predictions:

1. **TFLN interferometry:** At optical intensities $> 10^{21} \text{ W/m}^2$, discrete phase jumps following the Fibonacci hierarchy should occur:

$$\Delta\phi_{\text{discrete}} = 2\pi \cdot \phi^{-k} \quad (k \in \mathbb{N}) \quad (31)$$

2. **B-meson analysis:** The angular observables $\mathcal{K}_{1c}$ and $\mathcal{K}_{2ss}$ from [5] should deviate from SM predictions at higher statistics ($> 5\sigma$):

$$\mathcal{K}_{1c}^{\text{FFGFT}} = \mathcal{K}_{1c}^{\text{SM}} \cdot (1 + \alpha_{\text{FFGFT}} \cdot F_{\text{Fibonacci}}) \quad (32)$$

3. **Sub-Planckian resonances:** In nonlinear photonic crystals, modulation frequencies at multiples of $\xi \cdot \phi^{-k}$ should be detectable.

.12 Discussion: From Simulation to Hardware Transduction

The implication of the FFGFT is radical: photonic analogue computers based on TFLN do not merely simulate physical systems – they directly read out the clock of the spacetime hardware. The measured B-meson anomalies are the first experimental signatures of this transduction.

The remaining challenge is technological: the required laser intensity of $> 10^{32} \text{ W/m}^2$ ([7]) still lies several orders of magnitude above current petawatt systems ($\sim 10^{27} \text{ W/m}^2$). *(Note of 30 Sept. 2026: The prediction on TFLN interferometry above gives a threshold of $> 10^{21} \text{ W/m}^2$, whereas here $> 10^{32} \text{ W/m}^2$ is stated as required; the two statements are not compatible.)* Advances in TFLN fabrication are however promising intermediate steps.

.13 Conclusion

The **Fundamental Fractal Geometric Field Theory (FFGFT)** shows:

- Spacetime is based on a fundamental clock $\xi = 7500/t_P \approx 1.39 \cdot 10^{47} \text{ Hz}$.
- The 4D phase torus is not to be confused with a 3D torus plus time continuum; the time dimension is integrated as a fourth geometric dimension into the torus topology.
- Fractal feedback enforces phase-locking of the fundamental frequency, which stabilises matter.
- The 4D vector space is translated into a multidimensional matrix space via the matrix translation $\mathcal{T}$.

- Matrix multiplication corresponds physically to a 4D torus convolution.
- TFLN photonics achieves the required bandwidth and programmable complexity.
- LHCb data shows $\sim 4\sigma$ deviations from the SM – predictively consistent with the FFGFT.

The coming years will show whether at 5σ a new understanding of the spacetime hardware begins.

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